AP BIOLOGY TOPIC 1.6

Nucleic Acids: DNA, RNA, Directionality & Base Pairing

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

IST-1.A | 5′ → 3′ DIRECTIONALITY | ANTIPARALLEL & CHARGAFF

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.6 (NUCLEIC ACIDS: DNA, RNA, DIRECTIONALITY & BASE PAIRING)

CED Core Vocabulary

3 Core Components of Every Nucleotide — 5-Carbon Pentose Sugar: Deoxyribose (in DNA): Has a hydrogen (–H) on the 2′ carbon → chemically more stable, resists alkaline hydrolysis.

Ribose (in RNA) — Has a reactive hydroxyl (–OH) on the 2′ carbon → less stable, prone to self-cleavage.

Purines (2 fused C–N rings) — Adenine (A) and Guanine (G) — "Pure As Gold = 2 rings."

Pyrimidines (1 single C–N ring) — Cytosine (C) , Thymine (T, in DNA only) , and Uracil (U, in RNA only) — "CUT the Py = 1 ring."

Roles Beyond Genetic Storage — ATP (Adenosine Triphosphate): Ribonucleotide with 3 phosphate groups used for cellular energy transfer.

Signaling & Coenzymes — cAMP (second messenger), NAD^+ / FAD (electron carriers), and GTP .

Phosphodiester Linkage (Covalent) — Dehydration synthesis links the 5′ phosphate group of an incoming nucleotide to the 3′ hydroxyl (–OH) group of the last nucleotide in the chain.

Linear Strand Polarity (Directionality) — 5′ End: Terminates with a free phosphate group ($–PO_4^{2–}$) on the 5′ carbon.

§2. NUCLEOTIDE STRUCTURE (MONOMER OF DNA & RNA)

C, H, O, N, P

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Nucleotide Structure (Monomer of DNA & RNA) ➔ **3 Core Components of Every Nucleotide** ➔ **Roles Beyond Genetic Storage**

- **3 Core Components of Every Nucleotide:**
  - **5-Carbon Pentose Sugar:**
    - **Deoxyribose (in DNA):** Has a hydrogen (**–H**) on the **2′** carbon → chemically more stable, resists alkaline hydrolysis.
    - **Ribose (in RNA):** Has a reactive hydroxyl (**–OH**) on the **2′** carbon → less stable, prone to self-cleavage.
  - **Phosphate Group ($–PO_4^{2–}$):** Attached to the **5′** carbon of the sugar; imparts a negative charge and acidity.
  - **Nitrogenous Base (Attached to 1′ carbon):**
    - **Purines (2 fused C–N rings):** **Adenine (A)** and **Guanine (G)** — *"Pure As Gold = 2 rings."*
    - **Pyrimidines (1 single C–N ring):** **Cytosine (C)**, **Thymine (T, in DNA only)**, and **Uracil (U, in RNA only)** — *"CUT the Py = 1 ring."*
- **Roles Beyond Genetic Storage:**
  - **ATP (Adenosine Triphosphate):** Ribonucleotide with 3 phosphate groups used for cellular energy transfer.
  - **Signaling & Coenzymes:** **cAMP** (second messenger), **NAD^+ / FAD** (electron carriers), and **GTP**.

[CORE AXIOM / REMEMBER] **Nucleotide = 3-part unit** (5C Pentose Sugar + 5′ Phosphate Group + 1′ Nitrogenous Base).

§4. ANTIPARALLEL DOUBLE HELIX & CHARGAFF'S RULES

A=T (2) • G≡C (3)

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Antiparallel Double Helix & Chargaff's Rules ➔ **Antiparallel Double Helix** ➔ **Why Base Pairing Matters**

- **Antiparallel Double Helix:** Two polynucleotide strands coil around a central axis running in **opposite directions** (**5′ → 3′** paired alongside **3′ → 5′**).
- **Complementary Base Pairing (Purine always pairs with Pyrimidine → uniform ~2 nm diameter):**
  - **DNA Base Pairing:** **A–T** forms **2 hydrogen bonds**; **G–C** forms **3 hydrogen bonds**.
  - **RNA Base Pairing:** **A–U** (2 H-bonds) and **G–C** (3 H-bonds).
- **Why Base Pairing Matters:** Enables accurate **template-based DNA replication**, **transcription** (DNA → mRNA), and **translation** (codon–anticodon pairing).
- **Chargaff's Rules (Applies to Double-Stranded DNA):**
  - In any double-stranded DNA sample: **%A = %T** and **%G = %C** (total Purines **A+G = 50%** = total Pyrimidines **C+T = 50%**).
  - *Example Calculation:* If dsDNA is **18% Adenine → 18% Thymine** (36% A+T total) → remaining 64% is G+C → **32% Guanine & 32% Cytosine**.
  - **Single-stranded RNA** does *NOT* obey Chargaff's rules (unless folded into base-paired hairpins).

[MECHANISTIC NOTE] **Thermal Stability & G–C Content:** Because **C–G pairs form 3 H-bonds** vs. 2 H-bonds in A–T pairs, DNA regions with a higher **GC percentage are harder to denature** (require a higher melting temperature, T_m, to separate strands).

[AP EXAM FRQ SCORING STRATEGY] For full FRQ credit on DNA structure questions, always explicitly state **BOTH** terms: **1) Antiparallel strands (5′→3′ / 3′→5′)** AND **2) Complementary hydrogen-bonded base pairing (A–T, G–C)**.

Support this site
Scan to donate

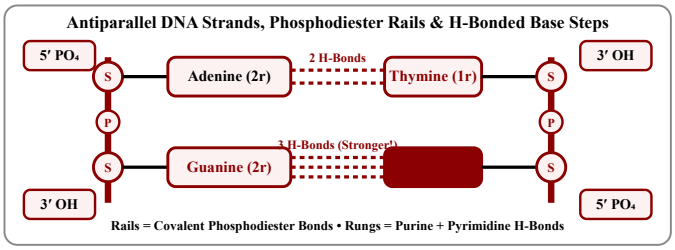
§3. SUGAR-PHOSPHATE BACKBONE & 5′ → 3′ DIRECTIONALITY

Phosphodiester Bonds

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Sugar-Phosphate Backbone & 5′ → 3′ Directionality ➔ **Phosphodiester Linkage (Covalent)** ➔ **Linear Strand Polarity (Directionality)**

- **Phosphodiester Linkage (Covalent):** Dehydration synthesis links the **5′ phosphate group** of an incoming nucleotide to the **3′ hydroxyl (–OH) group** of the last nucleotide in the chain.
- **Linear Strand Polarity (Directionality):**
  - **5′ End:** Terminates with a free phosphate group (**$-PO_4^{2-}$**) on the 5′ carbon.
  - **3′ End:** Terminates with a free hydroxyl group (**–OH**) on the 3′ carbon.
- **Rule of Nucleic Acid Synthesis (DNA Pol & RNA Pol):**
  - Strands **ALWAYS grow in the 5′ → 3′ direction**!
  - Incoming nucleoside triphosphates (dNTPs/NTPs) are **always added to the free 3′-OH end** (cleaving pyrophosphate **PP_i** provides the energy for polymerization).



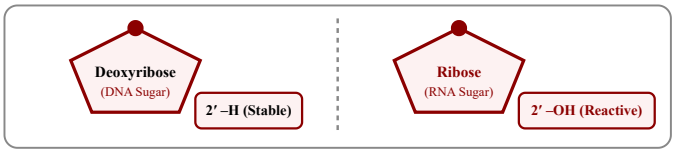
[COMMON AP EXAM FRQ TRAP] **Never mix up backbone vs. rung bonds:**
• **Sugar-phosphate backbone** = joined by strong **covalent phosphodiester bonds**.
• **Complementary base pairs** = joined by weak **hydrogen bonds** (so helicase can unzip strands during replication/transcription without breaking the genetic code!).

[CORE AXIOM / REMEMBER] **Backbone = ladder rails** (structural stability); **Bases = information steps** (sequence of bases = genetic code).

§5. DNA VS. RNA COMPARISON

Storage vs. Expression

FEATURE	DNA (DEOXYRIBONUCLEIC ACID)	RNA (RIBONUCLEIC ACID)
Pentose Sugar	**Deoxyribose** (**2′ –H**, lacks O)	**Ribose** (**2′ –OH** hydroxyl group)
Nitrogenous Bases	A, G, C, **Thymine (T)**	A, G, C, **Uracil (U)**
Strandedness	**Double-stranded** (antiparallel helix)	**Single-stranded** (can fold into 3D shapes)
Chemical Stability	**More stable** (2′-H + double helix)	**Less stable** (2′-OH susceptible to hydrolysis)
Biological Function	**Long-term hereditary storage**	**Short-term gene expression / catalysis** (mRNA, tRNA, rRNA, ribozymes)



[CORE AXIOM / REMEMBER] **DNA = permanent hard drive** (stable archive in the nucleus); **RNA = temporary working file** (disposable photocopy used for protein synthesis).

[MECHANISTIC NOTE] **AP CED Scope Note:** Memorizing the exact atomic ring structures of individual purine/pyrimidine bases is not required on the AP Exam — know which bases are **purines (A, G — 2 rings)** vs. **pyrimidines (C, T, U — 1 ring)** and their H-bond counts!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.6 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 1.6: Nucleic Acids: DNA, RNA, Directionality & Base Pairing ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, (n − 1) covalent bonds form and (n − 1) H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.6 to prove why the claim holds.

Credits: mitali120501@gmail.com